

Evaluating Digital Demand Signals and Territorial Mobility Patterns to Support Airline Seat Allocation

International Outbound Tourism Analysis in Spain (2019-2025)

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Abstract

This study evaluates whether digital demand signals can support airline seat allocation and outbound tourism planning decisions in Spain. The project combines the INE experimental outbound mobility dataset, DataForSEO monthly search volumes, Google Trends indices and a semantic destination-mapping layer created through web scraping and manual curation. Results revealed weak-to-moderate relationships under a simplified search-query architecture. Only 6 of 45 commercial destinations maintained useful correlations (Spearman $r \geq 0.3$) at the 11-month planning horizon relevant for airline allotment decisions. Monthly search volumes consistently outperformed Google Trends, which suffered from structural sparsity caused by relative normalization. Rather than validating a universal forecasting framework, the project evolved into a tourism intelligence framework integrating digital demand signals, territorial mobility patterns and seasonality analysis.

1. Introduction

Many travel agencies and tour operators negotiate flight seat allotments with airlines approximately one year in advance in order to secure availability and competitive package prices. This business model involves substantial financial exposure through deposits, contractual commitments and the operational risk of unsold inventory. Improving demand anticipation may reduce financial risk and support strategic tourism product planning.

The initial objective of this project was to evaluate whether digital search demand signals could improve tourism demand forecasting. However, the project progressively evolved toward a broader tourism intelligence framework after identifying substantial methodological limitations in public search-demand datasets.

2. Research Questions

- Can digital search behavior anticipate outbound tourism demand?

- Can territorial mobility datasets generate operational tourism intelligence for tour operators?

3. Data Sources

Source	Purpose	Granularity	Main limitations
INE experimental tourism mobility dataset	Observed outbound tourism flows	City-Province-country-month	No travel-purpose segmentation
DataForSEO monthly searches	Absolute demand signal	Keyword-month	No territorial granularity
Google Trends / Pytrends	Relative search interest	Keyword-month	Sparse and noisy signal
Tourism web scraping	Keyword enrichment	Destination terms	Manual semantic mapping required

A key methodological consideration is that weak correlations between digital demand signals and tourism flows cannot be attributed exclusively to the failure of search signals themselves. The INE experimental mobility dataset contains known structural limitations including partial mobile coverage, exclusion of children, eSIM and airplane mode underdetection, and aggregation corrections based on telecom market share. Simultaneously, the demand-signal framework relied on simplified tourism keywords such as 'viaje a + destination', which may insufficiently represent the complexity of real tourism planning behavior.

4. Methodology

- Extraction and cleaning of INE outbound tourism mobility datasets.
- Web scraping of tourism websites using Selenium and BeautifulSoup to build demand-signal keywords.
- Construction of tourism semantic keyword mapping layer.
- Pytrends anchor-based extraction pipeline.
- Monthly search volume extraction with DataForSEO.
- Geographic enrichment of provinces, autonomous communities and destinations.
- Territorial seasonality analysis and mobility concentration analysis.
- Continent-level behavioral adjustment factors derived from INE ETR leisure/total trip ratios, applied per continent and year to reduce non-leisure travel noise.

5. Main Results

5.1 Digital demand signal evaluation

The analysis revealed selective but limited predictive capacity. All correlations were computed using Spearman rank correlation, as Shapiro–Wilk normality tests confirmed non-normal distributions for `total_tourists` ($W=0.33$, $p\approx 0$), `monthly_searches` ($W=0.71$, $p\approx 0$) and `trend_index` ($W=0.86$, $p\approx 0$). Across 45 commercial destinations, the median Spearman r was 0.187 for monthly search volume and 0.176 for Google Trends. 40% of destinations showed statistically significant correlations for monthly searches ($p<0.05$) versus 30% for Google Trends. At the 11-month planning horizon, only 6 of 45 destinations (13%) maintained a useful Spearman $r \geq 0.3$: Cambodia, China, Brazil, Oman, Colombia and Cabo Verde. Search signals behave predominantly as concurrent indicators (optimal lag 0–3 months) rather than as annual forecasting tools. Google Trends datasets produced approximately 25% structural zero values due to relative normalization compression, explaining their systematically weaker performance compared to DataForSEO monthly search volumes.

Continent-level behavioral adjustment factors derived from INE ETR leisure/total trip ratios (América: 38%, Asia & Oceania: 61%, Africa: 30%), applied to reduce structural noise from non-leisure travel components. The improvement was weak and was not included in the final analysis.

5.2 Territorial outbound tourism intelligence

Strong territorial heterogeneity was identified across Spanish regions. Heatmaps and seasonality analyses revealed substantially different outbound tourism structures depending on province and autonomous community.

Outbound tourism outside Europe showed a strong Pareto concentration structure: Madrid and Barcelona together generate 66% of Spain's commercial long-haul outbound demand, and the top 7 of 50 provinces (14%) account for 80% of total volume. This territorial concentration has direct implications for seat allocation strategy, as demand signals from these provinces are structurally more reliable and representative.

6. Discussion

Several factors may contribute to the weak correlations observed throughout this analysis, including the experimental nature of the mobility dataset, simplified keyword selection, multilingual search behaviour, destination-specific planning horizons and heterogeneous travel motivations. Weak correlations should therefore not be interpreted as evidence that digital demand signals lack value, but rather as evidence that the relationship between search behaviour and tourism demand is more complex than initially hypothesised.

Additionally, digital search behavior is territorially heterogeneous and cannot be assumed to behave uniformly across Spain. The project demonstrates the importance of territorial analysis and semantic contextualization in tourism demand intelligence.

7. Conclusions

- Digital search signals showed selective predictive value: 40% of commercial destinations showed statistically significant Spearman correlations ($p < 0.05$), but only 13% (6 of 45) maintained useful signal ($r \geq 0.3$) at the 11-month planning horizon. Destinations with actionable predictive signal: Cambodia, China, Brazil, Oman, Colombia and Cabo Verde. For the remaining 87% of destinations, signals are predominantly concurrent (lag 0–3 months) and are better suited as real-time demand monitors than as annual forecasting tools.
- Monthly search volume (DataForSEO) was structurally more reliable than Google Trends: median Spearman $r = 0.187$ vs 0.176 , and 93% vs 89% positive correlations within the commercial subset. Google Trends produced approximately 25% structural zero values due to relative normalization compression, limiting its utility as a standalone demand signal.
- Semantic keyword mapping constitutes a critical methodological component for tourism intelligence.
- Tourism mobility datasets remain highly valuable for territorial tourism intelligence.

8. NEXT DEVELOPMENT STAGES

- Explore alternatives or other variables to undertake destination-level forecasting with ML.
- Integration of future airline capacity datasets (OAG / IATA).
- Improvement of detailed interactive dashboards and territorial flow maps.
- Incorporation of real agency or tour operator booking and occupancy data.

9. References

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Analytical scope: long-haul agency-driven markets



■ Not in scope — core Europe, border mobility, VFR demand

■ All destinations

All outbound trips from Spain, regardless of trip type or distance.

Includes:

- Short-haul Europe
(France, Portugal, Italy...)
- Morocco
(high VFR & migration component)
- Border mobility
(La Jonquera, Irun, Tarifa...)
- VFR & business travel
- All destination types
(leisure, niche, low-volume)

■ Strategic subset

Long-haul, agency-mediated destinations only.

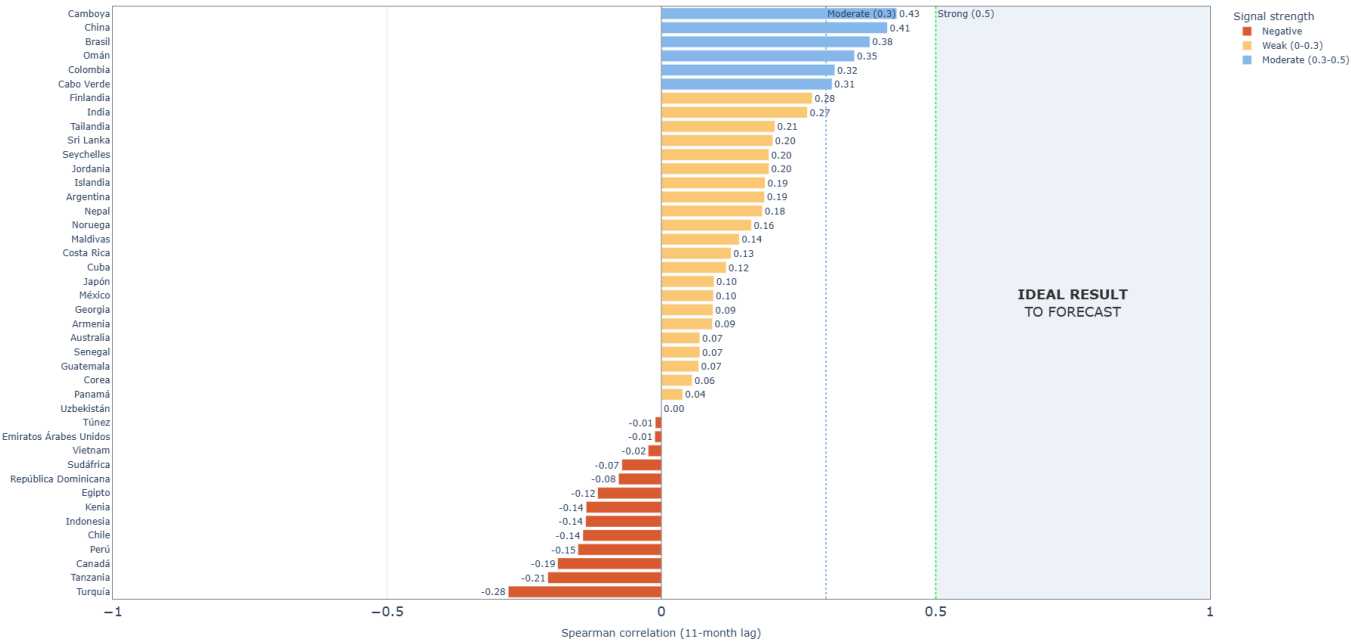
Excludes:

- ☒ Intra-European destinations
- ☒ Morocco
- ☒ USA

Keeps only:

- Medium & long-haul destinations
- High agency-dependency markets
- Air-dependent travel
- Package & organized tourism

Low correlation at lag 11 across 45 strategic destinations



Search-demand correlation at strategic lag windows (excl. COVID 2020-2021)

